

**PREPARATION AND VALIDATION OF A
GFEM**

Part: **2.2 – GLOBAL FINITE ELEMENTS
MODELS**

Chapter: **2210**

Title: **PREPARATION AND VALIDATION OF A
GFEM**

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1 INTRODUCTION

1.1 METHOD-SUMMARY

As mentioned in chapter 2199, Finite Elements Methods applied to Structural Analysis is based on dividing the structure in small pieces called 'Elements' in which we can define a simplified behaviour based on Partial Differential equations such as Linear elasticity.

The Global Finite Element Model (GFEM) is a Linear Structural FEM devoted to obtain an accurate representation of the internal loads at the different structural elements under the different types of loads acting on the structure.

In order to achieve this objective:

- The GFEM must be able to represent the external loads acting on the structure and the boundary conditions
- The GFEM must predict accurately the Load path

In Airbus, GFEM is created and analysed using MSC.NASTRAN FEA Code

1.2 SCOPE AND LIMITATIONS

The objective of this chapter is to define the main steps to develop, verify and analyze a Global FEM (GFEM) to be used in structural analysis process. The detailed description of the whole process is included in the references.

2 METHOD DESCRIPTION

The Global Finite Element Model (GFEM) is one relevant component of the Structural Justification process. Due to that, GFEM has Consolidated Modelling and Simulation rules (Ref. 1) and a Verification and Validation defined process (see Ref. 3). GFEM generation process inputs and outputs are also defined in BMS (see Ref. 4).

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- **Units:** GFEM must have a coherent set of units, as specified in Ref. 1. Usual set of Units are Newtons (N) for forces, millimetres (mm) for distance and Kg (or Mg, Tons) for Mass.
- GFEM **Mesh** is created based on Master Geometry (MG) that represents the external surfaces of the components (see Ref. 4) and according to Structural Scheme: Spar, ribs and stringers plane definitions, hinge line position, Interface points definition according to ICD, etc. The mesh used in GFEM is coarse in order to limit the Degrees of Freedom (DOF) of the Model. To generate the Mesh, FEM pre-processors as Hyperworks or MSC.Patran can support to this task (see Ref. 1)
- The Material cards of the elements are defined according to Ref. 1. Material properties (E_{ii} , G_{ij} , ν_{ij}) are defined as linear static **material properties** at the corresponding temperature, using mean values (tension-compression).
- The **Element properties** shall be defined according to the type of structure represented, according to Ref. 1. The modeller must document and track every thickness, laminate or cross-sectional Area used to model each element.
- **Interfaces Modelling:** The interfaces modelling is quite relevant to obtain I/F loads. According to Ref. 1 there will be at least 2 nodes at each I/F point joined by a connector element
 - o Connector elements (as CBUSH or CELASx) will link the corresponding DOF linked by the structural interface. For instance, a hinge fitting in the hinge line will link only 2 translational DOF (the ones perpendicular to Hinge Line)
 - o At some interfaces an additional connection by MPC can be added to the interface. This MPC can be used to simulate the failure of the interface or the rotation of a movable surface (see Ref. 1).
- **Loads** introduced to the GFEM must be critical Load cases either Limit or Ultimate Loads or Fatigue Loads. There can be different types or sources of loads applied to GFEM that can be combined for stress or fatigue analysis
 - o Mechanical Load cases are provided by Loads and Aeroelasticity departments in the agreed format (LOADS Interface Model (LIM)): These loads are provided in NASTRAN FORCE and MOMENT cards applied to GFEM nodes (see Ref. 1).
 - o Pressurization Load: Fuselage or Cabin can be pressurized. Internal pressure loads are introduced in GFEM in the corresponding pressure domain by means of NASTRAN PLOADx or nodal FORCE cards. The resultant of the internal pressure cases must be null at A/C level
 - o Temperature loads: A temperature mapping or temperature increment can be added to the GFEM to represent critical Hot Day or Cold day environment. Please take note the range of application of the Material properties and corresponding CTE included in the model.
- **Boundary conditions:** The GFEM must be supported to be analysed with NASTRAN in Linear Static solution. The boundary conditions are modelled by SPC cards
 - o In the case of a full A/C GFEM, an isostatic support will be used: just 6 DOF not linear dependent
 - o In the case of a Component GFEM, the model will be supported such as minimum impact on the load path, for example, adding some additional part of the structure between support and analysed component

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- There are additional constraints that may be added to the GFEM due to rotational singularities and other, that must be controlled and added to the GFEM (see Ref. 1)
- **Model Verification:** According to Ref. 3, the GFEM must be verified to assure a quality level. Some verification forms shall be filled to assure model verification, such as Geometrical/meshing checks, static analysis quality, thermoelastic and dynamic consistency.
- **Model Analysis.** Once the GFEM has been prepared and verified, it can be analysed running the corresponding Load Cases using Linear Static Solution in MSC.NASTRAN
 - The user must verify that there are no FATAL errors in output files and the coherency of the results
 - There can be different type of outputs of the GFEM analysis
 - ASCII files as F06 or punch files
 - Binary as OP2 or hdf5
 - Specific outputs can be extracted to feed Stress tools as TIFON or DIANA databases to accelerate results postprocessing
- After a Full Scale Test (FST) or Fatigue Test the GFEM has to be **Validated**. This can be done according to by means of displacement and stress/strains comparison between GFEM and Test results for areas in which this comparison can be done (clean areas with small local stress gradients and geometric representativity, no ply drop-off or section changes)
- **Model Documentation:** All the GFEM must be documented included all the items mentioned in this chapter, including Verification Forms and supported load cases analysed

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3 APPENDICES

ABBREVIATIONS AND SYMBOLS

<i>Symbol</i>	<i>Units</i>	<i>Definition</i>
A/C		Aircraft
FEM		Finite Element Model or Method
FST		Full Scale Test
GFEM		Global Finite Element Model
LIM		Loads Interface Model
MG		Master Geometry
I/F		Interface
ICD		Interface Control Document
DOF		Degrees of Freedom
CTE		Expansion Thermal Coefficient
MPC		Multi-Point Constraint
SPC		Single point Constraint

REFERENCES

	<i>Document Reference</i>	<i>Issue</i>	<i>Title</i>
Ref. 1	TEAS-NT-1810005	A	Guidelines for Global Finite Element Modelling In MSC.NASTRAN.
Ref. 2	[Dominique Madier]	1	Practical Finite Element Analysis for Mechanical Engineers
Ref. 3	DPI-ADC-007	F	Global Finite Element Model Verification and Validation Procedure
Ref. 4	TA.DV.PH.N059.EN	1	Produce Global Finite Element Model

SOFTWARE

<i>Program</i>	<i>Version</i>	<i>Description</i>
MSC.Nastran	2013.1	Finite Element Analysis Solver (Hexagon)
Hyperworks	2022.1	FE Model Pre and Post Processor (Altair)

Table 1. *Software programs referenced in this chapter*

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RECORD OF REVISIONS

Revision Date	Authors	Change Reason	Pages/Sub- chapters affected
0.1 DD/MM/AAAA	Name	First issue	All pages

Table 2. *Record of Revisions: authors, change reasons and sections/pages affected*

APPROVAL SIGNATURES TABLE

Dpt. \ Site	Getafe	Manching
Stress Methods (TEPAP-TL4)	Name DD/MM/AAAA	Name DD/MM/AAAA

Table 3. *Approval signatures table for last revision of this chapter*